EE Department - Sharif University of Technology

# Stochastic Bandits

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# Outline


1. What are Bandits?

2. Bandit Algorithms
   - 2.1 Explore then Commit
   - 2.2 Upper Confidence Bound

# What are Bandits?

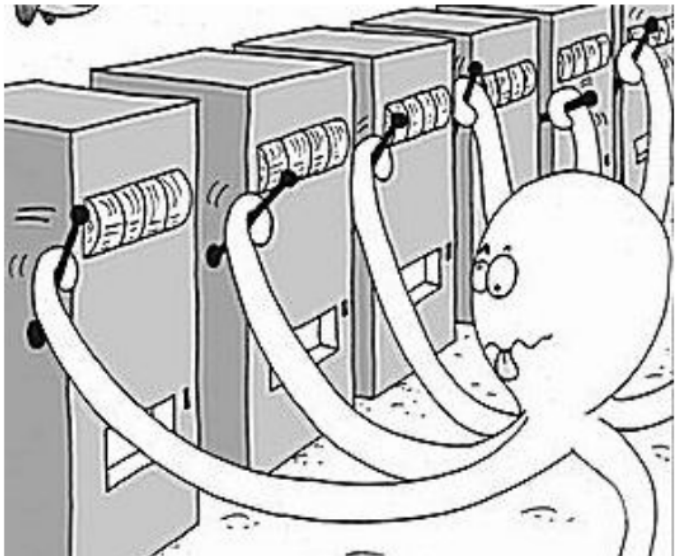
# What are Bandits?

| *Time* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Left arm* | $1 | $0 | | | $1 | $1 | $0 | | | | | |
| *Right arm* | | | $1 | $0 | | | | | | | | |

- **What are Bandits**?
  - Baby Reinforcement Learning.
  - Acting in the face of uncertainty.

# What are Bandits?

- A finite action set $\mathcal{A} = \{1, \ldots, k\}$
- For each $a \in \mathcal{A}$, there is an *unknown* distribution $\mathbb{P}_a$
- Learner chooses $A_t \in \mathcal{A}$ and observes $R_t \sim \mathbb{P}_{A_t}$
- **Goal**: Maximize $\sum_{t=1}^{n} R_t$



# Why Bandits?

- **Many applications**
  - Clinical trials/dose discovery
  - Advert placement
  - Ranking (eg., for search)
  - Dynamic pricing (eg., for Amazon products)
  - Isolating one interesting part of reinforcement learning
- **Exploration-vs-Exploitation**
- **Beautiful mathematics**

# Why Bandits?

- Let $\mu_a$ be the **mean** of $\mathbb{P}_a$ and $\mu^\star = \max_{a \in \mathcal{A}} \mu_a$.
- The **optimal action** $a^\star = \arg\max_{a \in \mathcal{A}} \mu_a$
- The task is to minimize **Regret**:

$$\text{Regret}_n = n\mu^\star - \mathbb{E}\Big[\sum_{t=1}^{n} R_t\Big]$$

  The price paid by the learner for not knowing $\mu$.

- To proceed, we need assumptions on the distributions:
  - Bernouli
  - Gaussian with unit variance
  - 1-subgaussian
  - Bounded
  - Many more ...

# Important Lemma

Let $\Delta_i = \mu^\star - \mu_i$ be the **suboptimality gap** for the $i$th arm and $T_i(n)$ be the number of times arm $i$ is played over all $n$ rounds:

## Lemma

$\text{Regret}_n = \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)]$

**Proof**:

$$\begin{aligned}
\text{Regret}_n &= n\mu^\star - \mathbb{E}\Big[\sum_{t=1}^{n} R_t\Big] = \mathbb{E}\Big[\sum_{t=1}^{n} (\mu^\star - R_t)\Big] \\
&= \mathbb{E}\Big[\sum_{t=1}^{n} \Delta_{A_t}\Big] = \mathbb{E}\Big[\sum_{t=1}^{n} \sum_{a \in \mathcal{A}} \mathbf{1}(A_t = a)\Delta_a\Big] \\
&= \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)],
\end{aligned}$$

which concludes the proof. $\square$

# Bandit Algorithms

# Simplest Policy: Explore then Commit

**Algorithm:** Explore then Commit

Choose each action $m$ times.
Find the *empirically* best action $I \in \{1, 2, \ldots, K\}$.
Choose $A_t = I$ for all remaining rounds.

# Simplest Policy: Explore then Commit

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## Theorem

Under the 1-subgaussian assumption of reward distributions, Explore then Commit, suffers

$$\text{Regret}_n \leq m\Delta + n\Delta \exp\Big(-\frac{m\Delta^2}{4}\Big).$$

- How to choose $m$, i.e., how much should be explore?
- We have a sharp transition from explore and exploit here. Is it possible to mix them?

# Analysis for $K = 2$

- Assume that $\mu_1 \geq \mu_2$. Let $\mu_i$ be the average reward after exploring.
- The algorithm commits to the wrong arm if

$$\hat{\mu}_2 \geq \hat{\mu}_1 \iff \hat{\mu}_2 - \mu_2 + \mu_1 - \hat{\mu}_1 \geq \Delta$$

- The random variable $\hat{\mu}_2 - \mu_2 + \mu_1 - \hat{\mu}_1$ is $\sqrt{\frac{2}{m}}$-subgaussian.
- **Regret bound**:

$$\begin{aligned}
\text{Regret}_n &= \mathbb{E}\Big[\sum_{t=1}^{n} \Delta_{A_t}\Big] = \mathbb{E}\Big[\sum_{t=1}^{2m} \Delta_{A_t}\Big] + \mathbb{E}\Big[\sum_{t=2m+1}^{n} \Delta_{A_t}\Big] \\
&= m\Delta + (n-2m)\Delta\mathbb{P}(\text{commit to the wrong arm}) \\
&\leq \underbrace{m\Delta}_{\text{Exploration}} + \underbrace{n\Delta \exp\Big(\frac{-m\Delta^2}{4}\Big)}_{\text{Exploitation}}.
\end{aligned}$$

# Analysis for $K = 2$

- **Regret bound**: $\text{Regret}_n \leq m\Delta + n\Delta \exp\left(\frac{-m\Delta^2}{4}\right)$.
- Bound minimized by $m = \frac{4}{\Delta^2} \log\left(\frac{n\Delta^2}{4}\right)$:

$$\text{Regret}_n \leq \Delta + \frac{4}{\Delta} \log\left(\frac{n\Delta^2}{4}\right) + \frac{4}{\Delta}.$$

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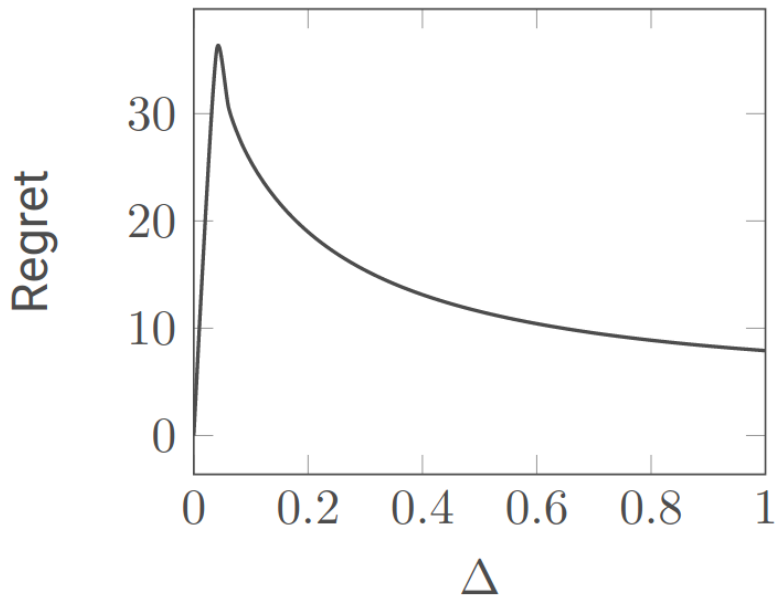

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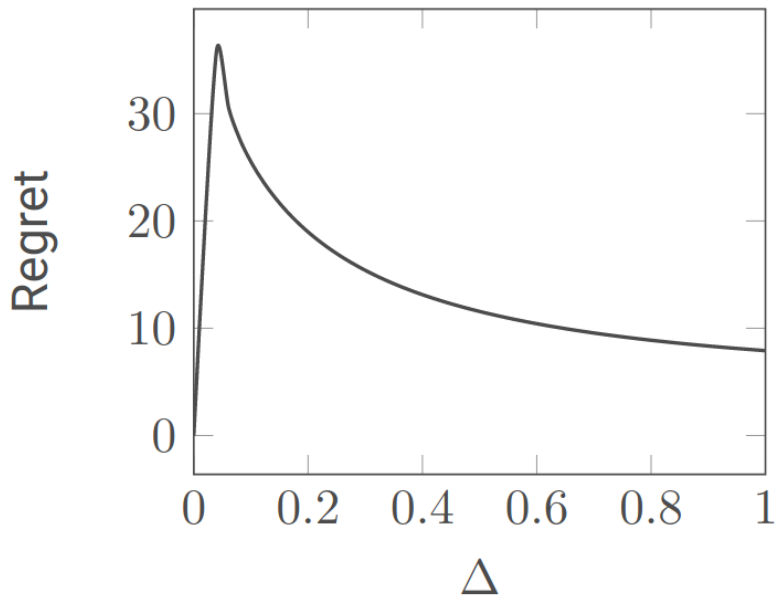


Worst case when $\Delta \approx \sqrt{\frac{1}{n}}$ with

$$\text{Regret}_n = C\sqrt{n}.$$

# Upper Confidence Bound

- First choose an action once.
- **Estimate**: $\hat{\mu}_i(t) = \frac{1}{T_i(t)} \sum_{s=1}^{t} \mathbf{1}(A_s = i) R_s$
- Optimistic estimate of mean = largest value it could plausibly be.
- Choose the action maximizing:

$$A_t = \arg\max_{i \in \mathcal{A}} \hat{\mu}_i(t-1) + \sqrt{\frac{2\log(t^3)}{T_i(t-1)}}$$

## Chernoff Bound

If $Z_1, \ldots, Z_n$ are independent $\sigma$-subgaussian, then $\mathbb{P}\left(\hat{\mu}_n \geq \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n}}\right) \leq \delta$.

## Regret Bound for UCB

If $Z_1, \ldots, Z_n$ are independent $\sigma$-subgaussian, then

$$\mathbb{P}\left(\hat{\mu}_n \geq \sqrt{\frac{2\sigma^2 \log(1/\delta)}{n}}\right) \leq \delta.$$

# Algorithmic Idea

- **Estimate** the mean of each arm
- Only play arms that are statistically plausibly optimal
- **Question**: What is this statistically plausible and which arm to play? $\implies$ We need assumptions: Gaussian with unit variance.